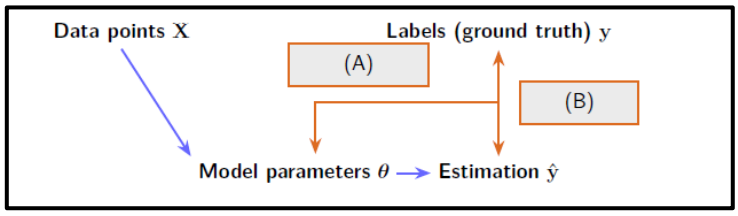
Task 3: Machine Learning Basics Foundations of Deep Learning

3(a) Understanding How We Obtain a Machine Learning Model

Question: The given figure illustrates how we obtain a machine learning model. Two labels are missing. We need to identify (A) and (B) and explain their functions.



Understanding the Process

In supervised machine learning, we provide the model with input data and the corresponding correct answers, called ground truth labels.

The learning process uses this information to find suitable model parameters. These parameters are then used to make predictions.

The main components are:

- **Data points (X):** The input features used to train the model.
- **Labels (y):** The actual correct answers corresponding to the input data.
- **Model parameters (θ):** The weights and biases that the model learns.
- **Estimation ($\hat{y}$):** The predictions produced by the trained model.

Missing Label (A): Learning Algorithm / Training Procedure

The learning algorithm uses the input data (X) and the corresponding ground truth labels (y) to learn suitable model parameters (θ).It adjusts the parameters during training so that the model produces predictions that are as close as possible to the actual labels.

Missing Label (B): Loss Function

The loss function measures the difference between the model's predictions ($\hat{y}$) and the actual ground truth labels (y).It calculates how much error the model makes. This error is used during training to improve the model parameters.

3(b) Identifying High Bias and High Variance

Question: A pedestrian detector is trained in four different settings. We assume that humans achieve an error of approximately 0%. For each case, we need to determine whether the model suffers from high bias, high variance, both or neither. We must also propose one concrete improvement for each problematic case.

Case	A	B	C	D
Training error	0.8 %	14 %	14 %	1.0 %
Validation error	12 %	15 %	29 %	1.6 %

Understanding the Concepts

1. Training Error

Training error measures how many mistakes the model makes on the data used to train it.

A low training error means that the model performs well on the training dataset.

2. Validation Error

Validation error measures how well the model performs on a separate dataset that was not used to train the model.

It helps us understand whether the model can generalise to new, unseen data.

3. Bias and Variance

High Bias:

High bias occurs when a model is too simple or has not learned the underlying patterns sufficiently. As a result, it performs poorly even on its training data. This is called *underfitting*.

High Variance:

High variance occurs when a model performs very well on training data but poorly on unseen data. This means that it has learned the training examples too specifically instead of learning general patterns. This is called *overfitting*.

Since the human error is approximately 0%, we can use it as a reference for understanding how much room there is for improvement.

Case A

Training Error: 0.8%

Validation Error: 12%

Analysis:

The training error is very low, which means the model has learned the training data well.

However, the validation error is significantly higher. This means that the model does not perform equally well on unseen data.

The large difference between training and validation errors indicates that the model is overfitting.

Diagnosis: High Variance.

Proposed Improvement:

Increase the amount and diversity of training data. More training examples can help the model learn general patterns instead of memorising the training examples.

Other possible measures include regularisation and data augmentation.

Case B

Training Error: 14%

Validation Error: 15%

Analysis:

Both training and validation errors are relatively high.

The errors are also very close to each other, which means the model performs similarly poorly on both datasets.

This indicates that the model has not learned the underlying patterns sufficiently.

Diagnosis: High Bias.

Proposed Improvement:

Use a more complex model with greater learning capacity so that it can learn more detailed patterns from the training data.

Other possible measures include improving the input features or reducing excessive regularisation.

Case C

Training Error: 14%

Validation Error: 29%

Analysis:

The training error is high, which indicates that the model is not learning the training data sufficiently. This is a sign of high bias.

The validation error is even higher, with a large difference between the training and validation errors. This indicates high variance as well.

Therefore, the model suffers from both underfitting and poor generalisation.

Diagnosis: High Bias and High Variance.

Proposed Improvement:

First, use a model with greater learning capacity to reduce high bias. Then, improve generalisation by adding more diverse training data and applying suitable regularisation.

The model should be evaluated again to ensure that both training and validation errors improve.

Case D

Training Error: 1.0%

Validation Error: 1.6%

Analysis:

Both training and validation errors are very low and close to each other.

This indicates that the model performs well on the training data and also generalises well to unseen data.

There is no significant indication of underfitting or overfitting.

Diagnosis: Neither High Bias nor High Variance.

Proposed Improvement:

No major improvement is necessary. The model is performing well, but its performance should continue to be monitored on new data.

3(c) Data Augmentation and Label Preservation

Question: Data augmentation is useful only if the transformation does not change the original label. We need to decide whether the following transformations are safe or unsafe and justify our answers.

What is Data Augmentation?

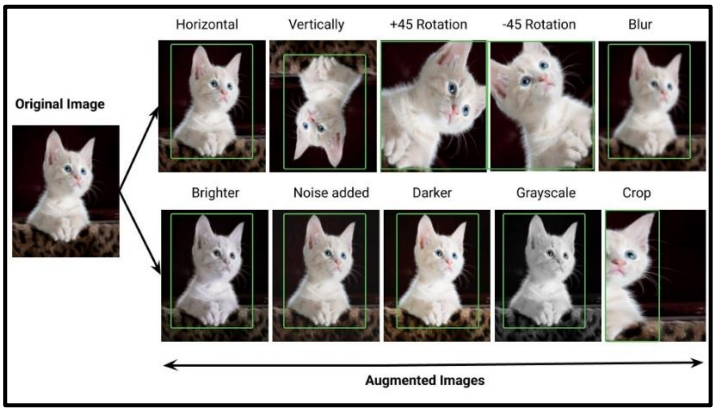
Data augmentation is a technique used to increase the diversity of training data by creating modified versions of existing examples.

For example, we can flip, rotate or adjust images to generate additional training examples.

It helps machine learning models learn more general patterns and can reduce overfitting.

However, there is one important rule: the transformation must preserve the original label. If the transformation changes the meaning of the image, it can teach the model incorrect information.

(i) Horizontal Flip – Cat vs. Dog



Decision: Safe

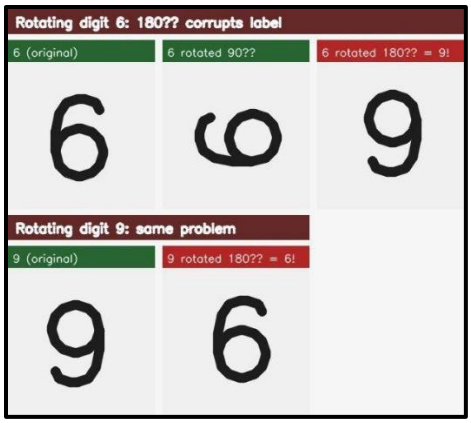
A horizontal flip reverses an image from left to right.

For example, if an image contains a cat, flipping it horizontally still produces an image of a cat. The same applies to a dog.

The transformation does not change the animal's identity or the classification label.

Therefore, horizontal flipping is a safe data augmentation technique for cat-versus-dog classification.

(ii) Rotation by 180 Degrees – Handwritten Digits



Decision: Unsafe

Rotating a handwritten digit by $180°$ can change its appearance and sometimes its identity.

For example, a handwritten 6 may look like a 9 after a $180°$ rotation.

The original label would still be 6, but the transformed image could be interpreted as 9.

This creates incorrect training information because the image and its label may no longer match.

Therefore, rotating handwritten digits by 180° is generally unsafe.

(iii) Conversion to Grayscale – Traffic Light State



Decision: Unsafe

Traffic light classification depends heavily on colour.

The three important states are:

- **Red:** Stop.
- **Yellow:** Caution.
- **Green:** Go.

Converting an image to grayscale removes its colour information. As a result, the model may no longer be able to distinguish between red, yellow and green traffic lights.

Since colour is essential for determining the correct label, grayscale conversion is unsafe for this particular classification task.